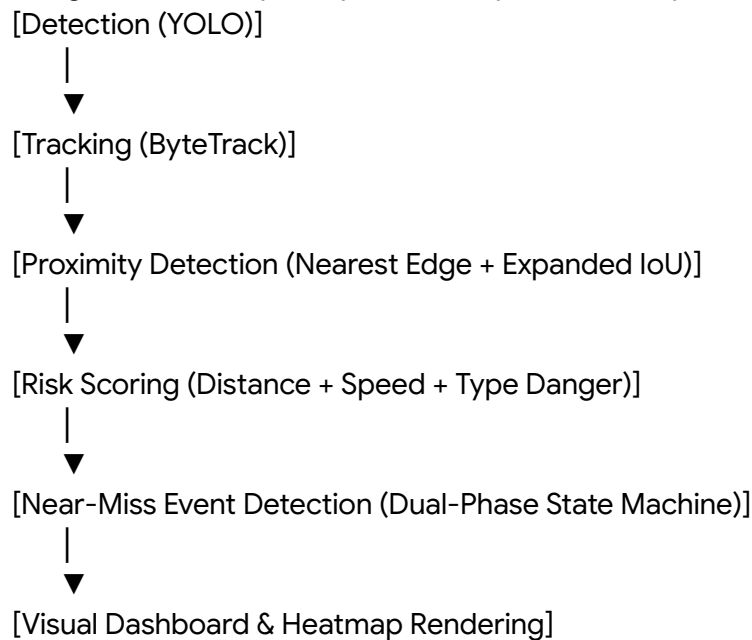


Project Report: Near-Miss Traffic Analytics (Smart City Road Safety System)

This report details the architectural and algorithmic design of the Near-Miss Traffic Analytics pipeline. Every threshold, constant, and mathematical design decision explained herein is directly derived from the accompanying Python implementation (traffic_near_miss.py).

1. System Overview

The system processes a video stream through a sequential six-stage edge-computing pipeline designed to identify, analyze, and map traffic safety hazards in real time:



1. **Object Detection:** Extracts spatial bounding boxes and class labels for active road users in each frame.
2. **Object Tracking:** Maintains identity consistency across frame boundaries, producing temporal motion histories.
3. **Proximity Detection:** Spatially filters pairs of tracked entities using a dynamic edge-to-edge distance and an expanded intersection-over-union metric.
4. **Risk Scoring:** Computes a dynamic risk score (0 to 100) for interacting pairs by modeling spatial overlap, closing speed, and class vulnerability.
5. **Near-Miss Event Detection:** Feeds continuous risk scores into a dual-phase state machine with recovery logic, cooldowns, and stationary noise filters to flag high-hazard

events.

6. **Visual Dashboard:** Renders safety metrics, historical risk profiles, trajectory projections, and localized hazard density maps directly onto the video feed.

2. Stage 1 — Object Detection

The object detection stage utilizes the **YOLO (You Only Look Once)** framework (specifically configured for standard deployments using yolo26m.pt or yolo26n.pt) to localize and categorize entities in the scene.

- **Confidence Threshold:** Set to args.conf with a default of **0.40** (**40%**). This provides an optimal balance between minimizing false-positive clutter in noisy urban environments and ensuring highly consistent bounding boxes.
- **Target Road User Classes:** Detection is restricted strictly to relevant road users defined by their COCO class IDs:
 - ID 0: person (Pedestrians/Vulnerable Road Users)
 - ID 1: bicycle
 - ID 2: car
 - ID 3: motorcycle
 - ID 5: bus
 - ID 7: truck

3. Stage 2 — Object Tracking

Temporal tracking is implemented using the **ByteTrack** algorithm, parameterized via bytetrack.yaml inside the YOLO framework.

- **Tracking History:** Center positions (c_x, c_y) are stored in a double-ended queue (deque) capped at a length of HISTORY_LEN = 30 frames.
- **Justification:** Storing **30** frames (equivalent to approximately **1** second of real-time footage at **30 fps**) provides a sufficient historical window to calculate smooth velocity vectors, perform speed estimation, filter out stationary environmental noise, and project future trajectories, all while maintaining a highly constrained memory footprint in an edge-computing environment.

4. Stage 3 — Proximity Detection

Proximity detection filters the massive combination of tracked objects down to active spatial interactions using two concurrent geometric criteria: **Nearest-Edge Distance** and **Expanded Intersection-over-Union (IoU)**.

Nearest-Edge Distance Computation

Unlike center-to-center distance, which penalizes large vehicles, the system computes the

exact minimum distance between the edges of bounding boxes A and B :

$$d_x = \max(0.0, \max(x_{1b} - x_{2a}, x_{1a} - x_{2b}))$$

$$d_y = \max(0.0, \max(y_{1b} - y_{2a}, y_{1a} - y_{2b}))$$

$$\text{dist} = \sqrt{d_x^2 + d_y^2}$$

Expanded Intersection-over-Union (IoU)

The standard bounding boxes are artificially expanded by $\text{IOU_EXPAND_SCALE} = 1.5$ (150%) around their centers before computing the overlap.

$$\text{center}_x = \frac{x_1 + x_2}{2}, \quad \text{center}_y = \frac{y_1 + y_2}{2}$$

$$w_{\text{expanded}} = (x_2 - x_1) \cdot 1.5, \quad h_{\text{expanded}} = (y_2 - y_1) \cdot 1.5$$

This step models a "safety envelope" around each road user. The expanded boxes must overlap, yielding an IoU greater than $\text{IOU_THRESHOLD} = 0.0$ for proximity verification.

Perspective and Scale Handling

To handle camera perspective (where distant objects appear smaller), the proximity threshold is scaled dynamically using the height of the smaller interacting object in the pair:

$$\text{edge_thresh} = \min(h_a, h_b) \cdot \text{EDGE_DIST_SCALE}$$

Where $\text{EDGE_DIST_SCALE} = 1.5$. If an interaction occurs in the background where box heights are small (e.g., 30 pixels), the threshold tightens to 45 pixels. In the foreground, where boxes are larger (e.g., 200 pixels), the threshold scales up to 300 pixels. This elegant formulation uses the object's relative scale as a proxy for depth.

Pair Filtering Logic

- **Excluded:** Person-person pairs (person ↔ person) are explicitly skipped. They do not pose a vehicular traffic safety hazard.

- **Included:** The pipeline requires that at least one member of the interacting pair is a vehicle (member of VEHICLE_CLASSES = {"car", "bus", "truck", "motorcycle", "bicycle"}).

5. Stage 4 — Risk Scoring

For every pair passing the proximity check, a risk score ranging from 0 to 100 is computed as the sum of three distinct safety risk vectors:

$$\text{Total Risk} = \min(100, \max(0, \text{Distance Score} + \text{Approach Speed Score} + \text{Type Danger Score}))$$

A. Distance Score (0 to 40 Points)

This component penalizes spatial proximity relative to the combined scale of the objects. The scale is defined by the sum of their bounding box diagonals:

$$\text{diag}_a = \sqrt{w_a^2 + h_a^2}, \quad \text{diag}_b = \sqrt{w_b^2 + h_b^2}$$

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$$\text{combined_diag} = \text{diag}_a + \text{diag}_b$$

Maximum Score (40): Triggered if the edge distance is less than half the combined diagonal ($\text{dist} < 0.5 \cdot \text{combined_diag}$).

- **Minimum Score (0):** Triggered if the edge distance exceeds double the combined diagonal ($\text{dist} > 2.0 \cdot \text{combined_diag}$).

- **Interpolated Score:** For intermediate distances, linear interpolation is applied:

$$\text{Distance Score} = 40 \cdot \frac{2.0 \cdot \text{combined_diag} - \text{dist}}{1.5 \cdot \text{combined_diag}}$$

B. Approach Speed Score (0 to 35 Points)

Velocity vectors (v_x, v_y) are calculated over the last $n = 10$ frames of tracking history. A unit direction vector (u_x, u_y) is drawn from Center A to Center B:

$$u_x = \frac{c_{xb} - c_{xa}}{d_{ab}}, \quad u_y = \frac{c_{yb} - c_{ya}}{d_{ab}}$$

The velocity vectors of both objects are projected onto this directional axis:

$$\text{proj}_a = v_{xa} \cdot u_x + v_{ya} \cdot u_y \quad (\text{positive if A moves toward B})$$

$$\text{proj}_b = v_{xb} \cdot (-u_x) + v_{yb} \cdot (-u_y) \quad (\text{positive if B moves toward A})$$

$$\text{closing_speed} = \text{proj}_a + \text{proj}_b$$

If the closing speed is positive (the objects are getting closer), the score scales linearly and is capped at **35**:

$$\text{Approach Score} = \min(35.0, \text{closing_speed} \cdot 2.0)$$

If they are moving apart, the score is **0.0**.

C. Type Danger Score (**0** to **25** Points)

Points are awarded based on the vulnerability and mass disparity of the interacting pair using the static mapping defined in TYPE_DANGER:

Interacting Pair	Weight	Justification
person ↔ any vehicle	25	Maximizes danger score for highly vulnerable road users exposed to any motorized/wheeled vehicle.
bicycle ↔ car/bus/truck/motorcycle	20	Captures vulnerable cyclists interacting with fast or heavy motorized vehicles.
motorcycle ↔ car/bus/truck	20	Models high-speed, low-protection motorists colliding with heavy transport.
vehicle ↔ vehicle (any type)	10	Standard collision penalty between protective metal-cabin vehicles.

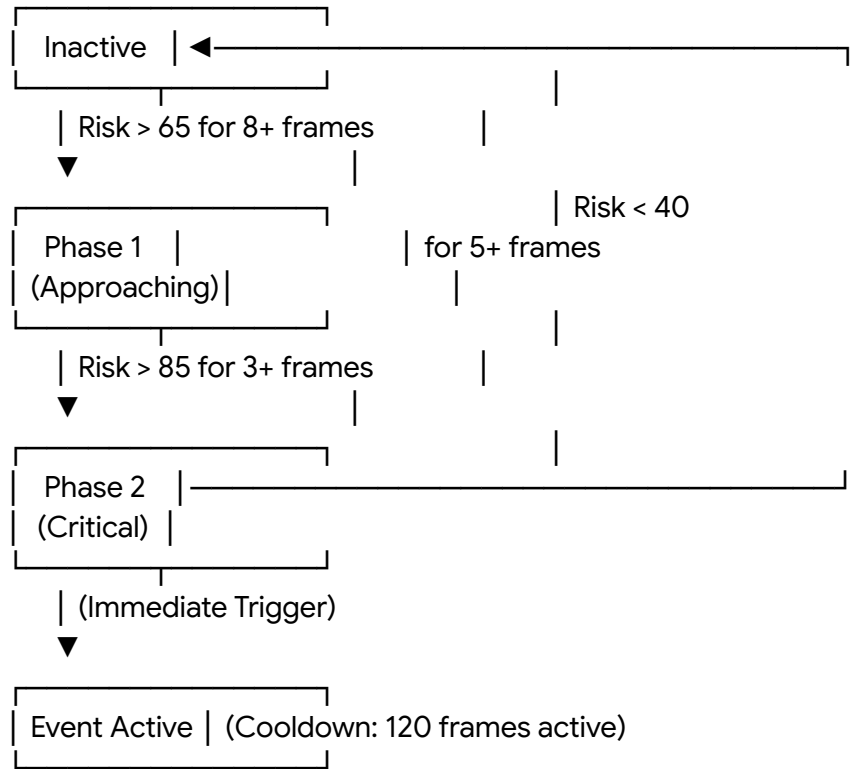
person ↔ person	5	Minimum safety hazard; virtually excluded by proximity logic.
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Risk Level Boundaries

- **High Risk:** $\text{Score} \geq 65$
- **Medium Risk:** $30 \leq \text{Score} < 65$
- **Low Risk:** $\text{Score} < 30$

6. Stage 5 — Near-Miss Event Detection

To prevent fleeting detection noise from polluting historical data, near-miss events are managed by a highly controlled, dual-phase state machine with temporal and spatial constraints:



State Machine Parameters & Guarded Scenarios

1. **PHASE1_THRESHOLD = 65 & PHASE1_FRAMES = 8:** Tracks sustained high risk. This guards against single-frame detector glitches or momentary tracking jumps that

artificially spike the risk score.

2. **PHASE2_THRESHOLD = 85 & PHASE2_FRAMES = 3**: Requires escalation to an extreme threat level. This ensures that steady-state vehicle interactions (like cars stopping naturally behind one another at a light with a risk score of 70) do not trigger a near-miss alert unless there is a physical emergency compression.
3. **RECOVERY_THRESHOLD = 40 & RECOVERY_FRAMES = 5**: The event remains active until the risk level consistently subsides. This prevents a single near-miss interaction from flickering on and off, treating it instead as one contiguous event.
4. **COOLDOWN_FRAMES = 120**: After an event is triggered, a 120 -frame cooldown (approx. 4 seconds) is applied to that specific pair. This prevents duplicate events from being registered as vehicles slowly clear the conflict zone.
5. **Stationary Filtering (is_stationary helper)**: Checks if an object's average velocity is less than $\text{STATIONARY_SPEED_THRESH} = 0.5$ pixels/frame over a window of $\text{STATIONARY_HISTORY_FRAMES} = 30$ frames. This guards against triggering near-miss alarms on vehicles parked close to pedestrian paths or static environmental obstructions.

7. Stage 6 — Visual Dashboard

The system's display utilizes consistent color coding and graphical layers to present complex spatial safety metrics on screen:

Color Coding (BGR Format)

- **COLOR_RED = (0, 0, 255)**: Active Near-Miss Event or Risk ≥ 85 .
- **COLOR_ORANGE = (0, 165, 255)**: High-Risk Interaction ($65 \leq \text{Risk} < 85$).
- **COLOR_YELLOW = (0, 255, 255)**: Medium/Low-Risk Interaction ($30 \leq \text{Risk} < 65$).
- **COLOR_GREEN = (0, 255, 0)**: Standard tracking status (no active interactions).

Visual Overlays

- **Distance Lines**: Rendered directly between the centers of interacting objects. The line color changes based on risk (Red, Orange, or Yellow) and is labeled with the precise edge distance in pixels (e.g., "142px").
- **Near-Miss Alerts**: When a near-miss triggers, a thick red bounding box is drawn encapsulating both objects. A warning banner (" $\triangle$ NEAR-MISS") is placed at the midpoint of their centers.

HUD Dashboard Elements

The top-left corner features a semi-transparent black overlay panel (320×140 pixels) at

70% opacity) containing:

1. **Near-Miss Count:** Total running count of validated near-miss events.
2. **Risk Level:** Current peak frame risk level text (High, Medium, Low) and numerical value.
3. **Vehicle Counter:** Active number of tracked vehicles in the scene.
4. **Timeline Bar:** A sliding horizontal timeline displaying the peak risk score of the last 100 frames. Individual frames are represented by vertical color-coded bars corresponding directly to their risk levels (Green, Yellow, Orange, Red).

8. Bonus Features Implemented

The codebase contains several high-performance safety analytics sub-systems that elevate it beyond basic object detection:

B1 — Dynamic Danger Heatmap

The system builds a spatial map representing hazard zones using 2D Gaussian accumulation:

- A float32 representation of the video dimension is updated in real time.
- During active near-misses, a 2D Gaussian blob is projected at the pair's midpoint using standard deviation HEATMAP_SIGMA = 20.0 and a crop radius of 3σ (60 pixels).
- To reflect changing road scenarios, the heatmap decays in each frame by HEATMAP_DECAY = 0.995 (0.5% loss per frame).
- The heatmap is colorized using the standard Jet colormap and blended over the output video frame at HEATMAP_ALPHA = 0.3 (30% opacity).

B2 — Camera Perspective Speed Estimation

Instead of relying on uncalibrated pixel movement, the code features a perspective projection model to calculate true ground speeds in km/h :

- **Camera Setup:** Mount height CAMERA_HEIGHT_M = 8.0 meters, field of view CAMERA_VFOV_DEG = 50.0 degrees, and tilt angle CAMERA_TILT_DEG = 30.0 degrees down from the horizontal.
- **Math Model:** The vertical normalized coordinate ($y_{\text{norm}} = y_{\text{center}} / H$) is translated to a ground ray angle:

$$\phi = \theta_{\text{tilt}} - \frac{\theta_{\text{vfov}}}{2} + y_{\text{norm}} \cdot \theta_{\text{vfov}}$$

$$\text{m_per_px} = \frac{H_{\text{camera}} \cdot \theta_{\text{vfov}}}{H_{\text{frame}} \cdot \sin^2(\phi)}$$

- This dynamically adjusts scale—providing high meters-per-pixel values near the horizon

(top of screen) and low values close to the camera (bottom of screen).

- The final displacement in meters is calculated over the last 10 frames of tracking, divided by the time delta (9 frames/fps), and multiplied by 3.6 to render the exact speed (e.g., "42.5 km/h") below each tracking label.

B3 — Trajectory and Collision Intersection Prediction

To predict collisions before they occur, the system includes a path prediction engine:

- **Position Projection:** Computes the mean velocity vector (v_x, v_y) of each object using its last 5 frames and projects its future location `TRAJECTORY_HORIZON = 15 frames` ahead (0.5 seconds).
- **Visual Trajectory:** Renders a yellow dashed directional arrow pointing from the object's center to its predicted future coordinates.
- **Intersection Check:** If two objects have an active risk score ≥ 65 , the system calculates the distance between their projected future positions. If this distance is less than `PREDICTED_INTERSECTION_THRESHOLD = 30 pixels`, a dashed magenta line is drawn connecting the predicted coordinates with a flashing overlay: "PREDICTED INTERSECTION".

B4 — Automated Safety Report Generation

Upon stream termination, the system aggregates all frame metrics to export a localized JSON safety report:

- **Intersection Risk Score:** Computed as the average of the top 5% highest peak risk scores in the entire run, providing an index of overall site danger.
- **Dangerous Window Detection:** Analyzes a rolling 30-second window (30 · fps frames) to isolate the exact start and end timestamps where the average risk score peaked.
- **Type Involvement:** Aggregates a histogram tallying which categories of road users were involved in logged near-misses.
- **Event Log:** Outputs an organized array of timestamps, participant tracking IDs, and exact peak risk scores.

9. Summary Table of Constants

Constant	Value	Role & Justification in the Pipeline

HISTORY_LEN	30	Maximum history length (in frames) kept in memory to track object centers.
DISPLAY_MAX_DIM	1000	Maximum window dimension constraint for resizing live video previews.
EDGE_DIST_SCALE	1.5	Height scale factor used to set the relative distance threshold for proximity checks.
IOU_EXPAND_SCALE	1.5	Expansion scale of bounding boxes to construct a safety envelope for IoU tests.
IOU_THRESHOLD	0.0	Minimum intersection over union required to validate proximity (any overlap).
CAMERA_HEIGHT_M	8.0	Real-world height of the physical camera in meters for perspective speed modeling.
CAMERA_VFOV_DEG	50.0	Vertical field of view of the camera lens in degrees for projection modeling.
CAMERA_TILT_DEG	30.0	Downward tilt angle of the camera in degrees from horizontal for perspective geometry.
TRAJECTORY_HORIZON	15	Number of frames projected into the future for linear path extrapolation.
PREDICTED_INTERSECTION_THRESHOLD	30	Maximum pixel distance between projected paths to trigger an intersection alert.

PHASE1_THRESHOLD	65	Risk threshold required to initiate Phase 1 (Approaching State) of a near-miss.
PHASE1_FRAMES	8	Consecutive frames above threshold required to complete Phase 1.
PHASE2_THRESHOLD	85	Risk threshold required to transition from Phase 1 to Phase 2 (Critical State).
PHASE2_FRAMES	3	Consecutive frames above Phase 2 threshold required to trigger a near-miss event.
RECOVERY_THRESHOLD	40	Risk score floor below which an ongoing near-miss event is considered resolved.
RECOVERY_FRAMES	5	Consecutive frames below the recovery floor required to terminate an active event.
COOLDOWN_FRAMES	120	Frames of system silence applied post-trigger to prevent repetitive near-miss events.
STATIONARY_SPEED_THRESHOLD	0.5	Maximum speed (pixels/frame) to classify an object as statically stationary.
STATIONARY_HISTORY_FRAMES	30	History window size used to evaluate if an object has remained stationary.
HEATMAP_SIGMA	20.0	Standard deviation (in

		pixels) of the Gaussian kernel added to near-miss hot spots.
HEATMAP_ALPHA	0.3	Blending weight (transparency) used to overlay the Jet colormap onto the output.
HEATMAP_DECAY	0.995	Multiplicative frame decay applied to the heatmap to fade historical hot spots.

10. Pipeline Limitations

The current implementation has several structural limitations based on its design assumptions:

1. **Strict 1D Perspective Assumption:** Speed estimation assumes that vertical position (y_{center}) is the sole indicator of distance. This assumes the ground plane is perfectly flat and objects travel parallel to the optical axis. Turn maneuvers, sideways motion, or changes in road elevation (slopes) will result in inaccurate speed calculations because horizontal displacement lacks a x -perspective scale.
2. **Linear Path Extrapolation:** The trajectory prediction system assumes linear motion (using a short 5-frame linear average). It cannot account for vehicular turning angles, acceleration, or deceleration at intersections. This results in false intersection predictions on curved roads and missed alerts at junctions where vehicles turn sharply.
3. **Fixed Pixel Thresholding:** Thresholds like `PREDICTED_INTERSECTION_THRESHOLD = 30` are set in flat pixel units. Due to camera perspective, 30 pixels in the foreground represents a few centimeters, while 30 pixels near the horizon represents several meters. This leads to disproportionately highly sensitive warnings for distant objects and late warnings for close objects.
4. **No Correction for Occlusions:** Tracking and calculations rely on 2D bounding boxes. Partial occlusions (e.g., a pedestrian walking behind a post or a car passing behind a truck) cause sudden shifts in box sizes and centers. This introduces instant noise into velocity estimation, triggering false approach-speed risk scores.